lec 20: Gaussian eli & LU

$L = L_1^{-1} L_2^{-1} \cdots L_{m-1}^{-1}$, $A = \underset{\text{lower}}{L}\,\underset{\text{upper}}{U}$

L_k: 将第 k 列, diag 下清空

$\vec{x}_k = \begin{bmatrix} x_{1k} \\ \vdots \\ x_{kk} \\ \vdots \\ x_{mk} \end{bmatrix} \xrightarrow{L_k} L_k \vec{x}_k = \begin{bmatrix} x_{1k} \\ \vdots \\ x_{kk} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$l_{jk} = \frac{x_{jk}}{x_{kk}} \ (k+1 \le j \le m)$ (if pivoting ⇒ 一定 ≤ 1) ∅

$L_k = I - l_k e_k^*$, $l_k = \begin{bmatrix} 0 \\ \vdots \\ l_{jk} \\ \vdots \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \\ -l_{k+1,k} \\ \vdots \\ -l_{m,k} \end{bmatrix}$

$L_k^{-1} L_{k+1}^{-1} = I + l_k e_k^* + l_{k+1} e_{k+1}^*$

$L = \begin{bmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & l_{32} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ l_{m1} & l_{m2} & \cdot & l_{m,m-1} & \end{bmatrix}$

algo $U=A$, $L=I$

for $k=1$ to $m-1$ 1~m-1 列

  for $j=k+1$ to m

    $l_{jk} = u_{jk}/u_{kk}$

    $u_{j,k:m} = u_{j,k:m} - l_{jk} u_{k,k:m}$

即: k+1 行及后, 每行 − 第 k 行 × 倍数

complexity: $\frac{2}{3}m^3$ flops

lec 21: pivoting

P, eg: $\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{bmatrix}$: 交换 row 3,4

$L_{m-1}P_{m-1} \cdots L_2 P_2 L_1 P_1 A = U$

$= \underbrace{(L'_{m-1} \cdots L'_2 L'_1)}\underbrace{(P_{m-1}\cdots P_2 P_1)}_{P} A$

$L'_k = P_{m-1} \cdots P_{k+1} L_k P_{k+1}^{-1} \cdots P_{m-1}^{-1}$ still lower tri

⇒ $PA = LU$

algo: partial pivoting

$U=A$, $L=I$, $P=I$

for $k=1$ to $m-1$

①找到 k 列 k^{th} 下, 绝对值最大元 其所在行

②交换 U row i ↔ row k

③交换 L 中已有的第 k 和 i 行前 k-1 列系数 (all)

④交换 P row i ↔ row k

for ... (LU消元)   (unstable)

$\rho = \frac{\max_{ij}|u_{ij}|}{\max_{ij}|a_{ij}|}$

不 pivot: $\tilde{L}\tilde{U} = A + \delta A$, $\frac{\|\delta A\|}{\|L\|\cdot\|U\|} = O(\epsilon_m)$

pivot: $\tilde{L}\tilde{U} = \tilde{P}A + \delta A$, $\frac{\|\delta A\|}{\|A\|} = O(\rho\,\epsilon_m)$ (backward stable)

意义: 和误差倍增正比, $\rho = 2^m$ ⇒ loss = O(m bits)

worst case for pivot: $\rho = 2^{m-1}$

lec 23 Cholesky factorize

if $A \in \mathbb{C}^{m\times m}$ hermitian p.d.

$X \in \mathbb{C}^{m\times n}$ rank full $(\min(m,n))$

⇒ $(X^*AX)^* = X^*AX$, 也 hermitian p.d.

⇒ A 的特征值都 $\in \mathbb{R}^+$

(反之: $\lambda_i \in \mathbb{R}^+\ \forall i$ ⇒ A p.d.)

$A = \begin{bmatrix} 1 & w^* \\ w & K \end{bmatrix}$

$\Rightarrow = \begin{bmatrix} 1 & 0 \\ w & I \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & K - ww^* \end{bmatrix}\begin{bmatrix} 1 & w^* \\ 0 & I \end{bmatrix}$

general A hermitian p.d.

$\Rightarrow A = \begin{bmatrix} a_{11} & w^* \\ w & K \end{bmatrix}$

$\alpha = \sqrt{a_{11}}$ ∅

$= \begin{bmatrix} \alpha & 0 \\ \frac{w}{\alpha} & I \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & K - \frac{ww^*}{a_{11}} \end{bmatrix}\begin{bmatrix} \alpha & \frac{w^*}{\alpha} \\ 0 & I \end{bmatrix}$

$= R_1^* A_1 R_1$ p.d

∅ any principal submatrix of p.d ⇒ also p.d.

$\Rightarrow A_1 = R_2^* A_2 R_2$

$\Rightarrow \cdots \Rightarrow A = \underbrace{(R_1^* R_2^* \cdots R_m^*)}_{R^*\ (\text{lower})}\underbrace{(R_m \cdots R_2 R_1)}_{R\ (\text{upper})}$

$A = R^*R$, $r_{jj} > 0$

algo Cholesky $A = LL^*$

for $i = 1, \ldots, n$:

对角线元素 $l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} |l_{ik}|^2}$ $= l_{ik}\bar{l}_{ik}$

for $j = i+1, \ldots, n$: i^{th} row 前面项平方和

算 i^{th} col 下方非diag元素: $l_{ji} = \frac{1}{l_{ii}}\left(a_{ji} - \sum_{k=1}^{i-1} l_{jk}\bar{l}_{ik}\right)$

Stability: $\tilde{R}^*\tilde{R} = A + \delta A$, $\frac{\|\delta A\|}{\|A\|} = O(\epsilon_m)$

To solve $Ax=b$: $(A+\Delta A)\hat{x} = b$, $\frac{\|\Delta A\|}{\|A\|} = O(\epsilon_m)$ backward stable (and well-conditioned)

complexity: $\frac{1}{3}n^3$ (快于LU一倍)

ex $A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$ $L_{11} = \sqrt{4} = 2$, $L_{21} = \frac{1}{2}(2-0) = 1$, $L_{22} = \sqrt{3-1^2} = \sqrt{2}$

note: $\|R\|_2 = \|R^*\|_2 = \|A\|_2^{\frac{1}{2}}$

note: any hermitian Hessenberg matrix 都是 tridiagonal 的

lec 24 eigen

review: FTA: $p_A(z) = \det(zI - A)$ (λ eigen iff $\lambda I - A$ singular)

$\Rightarrow p_A(z) = (z-\lambda_1)\cdots(z-\lambda_m)$

geo multi of λ = $\dim(E_\lambda)$ = null$(A - \lambda I)$

alge mult of λ = multi of λ in p_A

Thm alge m ≥ geo m

Thm X nonsing ⇒ A, $X^{-1}AX$ has same char poly and alg, geo (everything)

say A similar to B if $B = X^{-1}AX$ for nonsin X

Thm $\det A = \prod \lambda_i$, $\operatorname{tr}(A) = \sum \lambda_i$

undefective: means algebraic mul = geo mul iff A diagonalizable (存在 eigen decomp)

* 未必可逆

unitarily diagable: ∃ unitary Q s.t. $A = Q\Lambda Q^*$

Thm hermitian matrices are uni. diagable with real eigens

Thm A matrix is uni diagable iff it is normal $AA^* = A^*A$

Schur factorization: ∀ square mat A

$A = QTQ^*$ where Q uni, T upper tri.

method: $A \xrightarrow[\text{Hessenberg}]{\text{Reduction to}} H \xrightarrow[\text{iteration}]{\text{QR}} T$

lec 26 reduction to Hessenberg

$A \xrightarrow{Q_1^*\cdot} Q_1^*A \xrightarrow{\cdot Q_1} Q$

choose Householder reflector Q_1^* that leaves first row unchanged

自动成立, 因为 householder ⇒ Q_1 on right leaves first col unchanged

Q_k 只是 右下角 $(n-k)\times(n-k)$ ⇒ $\underbrace{Q_{m-2}^* \cdots Q_1^*}_{Q^*} A \underbrace{Q_1 \cdots Q_{m-2}}_{Q} = H$

$Q_k = \begin{bmatrix} I_k & 0 \\ 0 & \hat{Q}_k \end{bmatrix}$

complexity: $\frac{10}{3}m^3$ for 普通, $\frac{4}{3}m^3$ for Hermit

Stability: backward ✓ ($\tilde{Q}\tilde{H}\tilde{Q}^* = A + \delta A$, $\frac{\|\delta A\|}{\|A\|} = O(\epsilon_m)$)

$\lambda_{min} = \min_{\|x\|=1} r(x)$

$\lambda_{max} = \cdots$

lec 27 Rayleigh Quotient, Inverse iteration

$r(x) = \frac{x^TAx}{x^Tx} \Rightarrow \nabla r(x) = \frac{2}{x^Tx}(Ax - r(x)x)$

if $x \in E_\lambda \Rightarrow r(x) = \lambda$ ∅

if $\nabla r(x) = 0$, $x \ne 0$ ⇒ x is a eigen vec, and $r(x)$ is an eigen value

for any eigenvec q_j: $r(x) - r(q_j) = O(\|x - q_j\|^2)$ as $x \to q_j$

algo Power iteration

v_0 random unit vector

for $k \to \infty$

  $w = Av^{(k-1)}$

  $v^{(k)} = w/\|w\|$

  $\lambda^{(k)} = (v^{(k)})^T A v^{(k)}$ Rayleigh quotient

PM works since:

$v^{(0)} = a_1 q_1 + \dots + a_m q_m$

$v^{(k)} = c_k A^k v^{(0)} = c_k \lambda_1^k \left(a_1 q_1 + a_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k q_2 + \dots + a_m \left(\frac{\lambda_m}{\lambda_1}\right)^k q_m\right)$

$\|v^{(k)} - q_1\| = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right), \quad |\lambda^{(k)} - \lambda_1| = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^{2k}\right)$

algo inverse iteration (找到最接近μ的eigenvalue 的eigen vector)

$v^{(0)}$ random unit

for $k = 1, 2, \dots$

Solve $(A - \mu I)w = v^{(k-1)}$ for w

$v^{(k)} = w / \|w\|$

$\lambda^{(k)} = (v^{(k)})^T A v^{(k)}$ Rayleigh

Suppose λ_J closest to μ, λ_K second.

$\Rightarrow \|v^{(k)} - q_J\| = O\left(\left|\frac{\mu - \lambda_J}{\mu - \lambda_K}\right|^k\right)$

$|\lambda^{(k)} - \lambda_J| = O\left(\left|\frac{\mu - \lambda_J}{\mu - \lambda_K}\right|^{2k}\right)$

Alternating: (to get a pair)

$v^{(0)}$ random unit

$\lambda^{(0)} = (v^{(0)})^T A v^{(0)}$

for $k = 1, 2, \dots$

Solve $(A - \lambda^{(k-1)} I)w = v^{(k-1)}$ for w

$v^{(k)} = w / \|w\|$

$\lambda^{(k)} = (v^{(k)})^T A v^{(k)}$

$\Rightarrow \|v^{(k+1)} - q_J\| = O(\|v^{(k)} - q_J\|^3)$

$|\lambda^{(k+1)} - \lambda_J| = O(|\lambda^{(k)} - \lambda_J|^3)$

note:

$Av = \lambda v$

$\Rightarrow p(A)v = p(\lambda)v$

eigen vec 不变, value 随多项式

$Av = \lambda v$

$\Rightarrow (A - \mu I)v = (\lambda - \mu)v$

$\Rightarrow (A - \mu I)^{-1} v = \frac{1}{\lambda - \mu} v$

eigen vec 不变, value 变为 $\frac{1}{\lambda - \mu}$

$Av = \lambda v$

$\Rightarrow$ for $B^{-1}AB$ (similarity trans)

$(B^{-1}AB)w = \lambda (B^{-1}v) = \lambda w$

λ不变, eigen vec 变为 $B^{-1}v$

QR by household: bw stable

solve $Ax = b$ by household QR: bw stable

$(A + \Delta A)\tilde{x} = b$, $\frac{\|\Delta A\|}{\|A\|} = O(\varepsilon_m)$

and $\frac{\|\tilde{x} - x\|}{\|x\|} = O(\kappa(A)\varepsilon_m)$

Lec 31

$A = U\Sigma V^* \Rightarrow A^*A = V\Sigma^*\Sigma V^*$ (cov matrix)

$A^*U = V\Sigma^* = V\Sigma$

reduce SVD to an eigen problem:

$$\begin{bmatrix} 0 & A^* \\ A & 0 \end{bmatrix} \begin{bmatrix} V & V \\ U & -U \end{bmatrix} = \begin{bmatrix} V & V \\ U & -U \end{bmatrix} \begin{bmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{bmatrix}$$

(for square A)

Lec 28 QR

pure QR: $A^{(0)} = A$

每次 $A^{(k-1)}$ 做QR分解 $\Rightarrow = Q^{(k)}R^{(k)}$

然后 $A^{(k)} = R^{(k)}Q^{(k)}$

$(= Q^{(k)T} A^{(k-1)} Q^{(k)}$, by $Q^{(k)}$ being ortho)

if converge $A \to$ upper tri with diag the eigenvalues

Practical QR

1. 在iterate前, 用Householder reduce A to Hesenburg
2. 不用 $A^{(k)}$ 而改用 $A^{(k)} - \mu^{(k)} I$, $\mu^{(k)}$是个 eigenvalue estimation $A^{(k)}_{j,j+1}$
3. 一旦某个off diag元素足够接近0, 把它及斜下都由0, 得到 $A^{(k)} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix}$

Simultaneous iteration ($\Leftrightarrow$ pure QR)

Pick $\hat{Q}^{(0)} \in \mathbb{R}^{m \times n}$ with orthonormal cols

for $k = 1, 2, \dots$

$Z = A\hat{Q}^{(k-1)}$

$\hat{Q}^{(k)}\hat{R}^{(k)} = Z$

Simul iter: $\underline{Q}^{(0)} = I$

$Z = A\underline{Q}^{(k-1)}$

$Z = \underline{Q}^{(k)}R^{(k)}$

$A^{(k)} = (\underline{Q}^{(k)})^T A \underline{Q}^{(k)}$

unshifted QR: $A^{(0)} = A$

$A^{(k-1)} = Q^{(k)}R^{(k)}$

$A^{(k)} = R^{(k)}Q^{(k)}$

$\underline{Q}^{(k)} = Q^{(1)} \dots Q^{(k)}$

$\underline{R}^{(k)} = R^{(k)} \dots R^{(1)}$

Thm They generate identical seq of $\underline{R}^{(k)}, \underline{Q}^{(k)}, A^{(k)}$

且 $A^k = \underline{Q}^{(k)}\underline{R}^{(k)}$: 为 A^k 的QR分解

(since $A^k = A\underline{Q}^{(k-1)}\underline{R}^{(k-1)} = \underline{Q}^{(k)}\underline{R}^{(k)}$ by induction)

Lec 12

$\hat{\kappa} = \lim_{\delta \to 0} \sup_{\|\delta x\| \le \delta} \frac{\|\delta f\|}{\|\delta x\|}$

$= \sup_{\delta x} \frac{\|\delta f\|}{\|\delta x\|} = \|J(x)\|$

$\kappa = \lim_{\delta \to 0} \sup_{\|\delta x\| \le \delta} \left(\frac{\|\delta f\|}{\|f(x)\|} \Big/ \frac{\|\delta x\|}{\|x\|}\right)$

$= \sup_{\delta x} \sim = \frac{\|J(x)\|}{\|f(x)\| / \|x\|}$

eg $f(x) = x_1 - x_2$, $J = [1, -1]$

$\kappa = \frac{\|J\|}{\|f(x)\| / \|x\|} = \frac{2}{|x_1 - x_2| / \max\{|x_1|, |x_2|\}}$

condition number of a matrix:

$\kappa(A) = \|A\|\|A^{-1}\| \left(= \frac{\sigma_1}{\sigma_m} \text{ if 2-norm}\right)$

$\|A\| \frac{\|x\|}{\|b\|}, \|A^{-1}\| \frac{\|b\|}{\|x\|} \le \kappa(A)$

backward stability: $\tilde{f}(x) = f(\tilde{x})$

stability: $\frac{\|\tilde{f}(x) - f(\tilde{x})\|}{\|f(\tilde{x})\|} = O(\varepsilon_m)$

for some $\tilde{x}$ with $\frac{\|\tilde{x} - x\|}{\|x\|} = O(\varepsilon_m)$

Thm if f has cond number κ and $\tilde{f}$ is backward stable

$\Rightarrow \frac{\|\tilde{f}(x) - f(x)\|}{\|f(x)\|} = O(\kappa(x)\varepsilon_m)$

Thm if A real sym, eigens 不同, 则 $A^{(k)} \to \mathrm{diag}(\lambda_1, \dots, \lambda_m)$ linearly